

# Luohua Yang

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**Right to work in the UK: Graduate Route visa (2 years, no sponsorship required)** — application submitted on course completion; currently on Student visa until 31 Jan 2027 *[update once approved]*

## SUMMARY

MSc Statistical Data Science graduate (University of Birmingham, Sept 2026) comfortable owning complete data workflows — scheduled collection and storage, SQL analysis, BI dashboards (Tableau, ECharts), and LLM-assisted reporting. Statistics and finance background keeps analysis grounded in real business problems. Bilingual English/Chinese. Two production-grade personal projects run daily on GitHub Actions; architecture and review led end-to-end with AI pair programming (Claude Code).

## SELECTED PROJECTS

### stock-monitor — Automated Multi-market Equity Analysis & Daily Report System 2025–2026 · open-source MIT

- Automated daily collection and storage for 20+ equities across 4 markets (yfinance primary; AKShare/baostock A-share fallback; Tavily real-time news), cutting manual research time by ~90%.
- Built a configurable LLM analysis layer (multiple mainstream providers) scoring candidates with Buffett-value and Fisher-growth frameworks plus DCF valuation — surfaced 5+ potential opportunities daily.
- Cross-validated DCF growth assumptions across PEG-implied, consensus-estimate and historical sources (50% cap), preventing single-quarter earnings spikes from distorting valuations.
- Shipped a Flask + ECharts dashboard (candlesticks, dual-factor radar charts, margin-of-safety buy ranges) with Feishu card alerts on a GitHub Actions weekday cron; runtime data-integrity checks and per-report AI cost tracking (~\$0.003 with default model).
- Designed the architecture and reviewed all code (AI-assisted pair programming). [github.com/VerticalFall/stock-monitor](https://github.com/VerticalFall/stock-monitor)

### obsidian-vault — Automated News Knowledge Base & Bilingual Daily Briefing 2025–2026 · open-source

- Engineered a GitHub Actions pipeline aggregating 4 automated content lines (AI news radar, trend radar, tracked indie builders, X/Twitter feeds) into a searchable Markdown knowledge base.
- Designed a DeepSeek-powered 3-stage topic router distilling 100+ daily articles into a 5-minute bilingual (CN/EN) briefing — a production-style text classification/routing pipeline.
- Built 4 per-source renderers plus a health-check script (data-integrity and pipeline self-testing with alerts to a system-health log); 160+ editions produced automatically since July 2026 at zero infrastructure cost. [github.com/VerticalFall/obsidian-vault](https://github.com/VerticalFall/obsidian-vault)

## EXPERIENCE

### Data Operations Intern — Tungee (AI sales-intelligence SaaS, ~1,000 staff) Guangzhou · Jun–Sep 2024

- Replaced manual logging with a scheduled pipeline collecting daily KPIs (followers, views, comments) for 50+ KOL accounts on Douyin and Bilibili (10k+ data rows processed daily).
- Ran retention/funnel analyses on KOL accounts *[add object & key finding — one line]*; insights fed weekly content-strategy reviews.
- Built Tableau dashboards on performance trends and audience engagement, used in weekly content reviews.
- Turned data into content-strategy recommendations, helping the team identify top-performing content formats ~2× faster.

## EDUCATION

### University of Birmingham — MSc, Statistical Data Science 2025–2026

Dissertation submitted Sept 2026; results pending *[update when confirmed — e.g. Distinction]*

- Thesis *Generative vs. Discriminative Classifiers for Financial Sentiment Analysis*: benchmarked multinomial Naive Bayes vs linear SVM on Financial PhraseBank (4,846 expert-labelled financial news sentences, 3 classes), empirically testing Ng–Jordan learning-curve theory — generative lead with crossover at  $n \approx 256$ –512; SVM best at full scale (accuracy 0.757 vs 0.698; macro-F1 0.691 vs 0.515).
- Robustness experiments on label noise (annotator agreement) and class imbalance: recovered the rare negative class from recall 0.151 → 0.456 via class-prior/weight adjustment.
- Coursework: machine learning, deep learning, statistical modelling, econometrics.

### Guangdong University of Foreign Studies — BSc, Statistics 2021–2025

Probability, statistical inference, econometrics. Research project member: built and maintained a weekly incremental news crawler (requests, MySQL) for Southeast Asia news sources.

## SKILLS

**SQL** (joins, window functions, aggregations) · **Python** (pandas, NumPy, scikit-learn, PyTorch) · **Statistics & hypothesis testing** · **ML classification & sentiment analysis** · **BI**: Tableau, ECharts · **Pipelines**: GitHub Actions, cron, requests, MySQL, REST APIs · **LLM integration** (multi-provider, RAG basics) · Git, Docker, Flask, Linux

**Languages**: Mandarin (native) · English (professional working proficiency)